Vicente Andree Honorato Rodríguez

CONTACT INFORMATION

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EDUCATION

Universidad Técnica Federico Santa María (UTFSM), San Joaquín, Chile

Bachelor's degree in Astrophysics, 2021–2025

Universidad Técnica Federico Santa María – Pontificia Universidad Católica de Valparaíso (PUCV), Valparaíso, Chile

PhD Student, Doctoral Program in Physical Sciences, 2026–Present

RESEARCH AND TEACHING EXPERIENCE

UTFSM, Chile: Research Assistantships, November 2022–Present

From late 2022 to September 2025, I worked as a research assistant under the supervision of Prof. Antonio Montero-Dorta, focusing on the analysis of dark-matter halo density profiles using the high-resolution hydrodynamical cosmological simulation IllustrisTNG50. The project focused on the inner regions of these profiles and addressed questions related to the *core–cusp* dichotomy. I performed several profile fits based on modifications of the Navarro, Frenk & White (NFW) model. I then connected the parameters derived from these fits with physical properties of the galaxies hosted by the halos—such as star formation rate, color, and stellar mass—to explore the halo–galaxy connection. We also studied the impact of baryons on the profiles by comparing against the gravity-only (DM-only) version of the same simulation and by evaluating their evolution with redshift, which allowed us to identify trends and behaviors not previously reported in the literature. The resulting manuscript is currently available on arXiv and has been submitted to *Astronomy & Astrophysics*. The paper is titled *"The Inner Dark-Matter Structure of Galaxies"* (Vicente Honorato, Antonio D. Montero-Dorta, M. Celeste Artale, and Ankit Kumar, submitted, 2025).

In the second semester of 2025, I began a research assistantship with Prof. Matías Montesinos on *Tidal Disruption Events* (TDEs) in planet–star systems, focusing on hydrodynamical simulations and radiative post-processing to characterize their emission and establish diagnostics that may help distinguish them from other astrophysical channels through their light curves. Through comparison with analytical models and survey-oriented validation, I aim to connect simulation results with observations and develop identification criteria.

UTFSM, Chile: Teaching Assistantships, March 2023–Present

During the first semester of 2023, I served as a teaching assistant for *General Physics II* (FIS120, *electromagnetism*). In coordination with Prof. Vladimir Juricić, I supported students and graded exams.

In the second semester of 2024, I worked as a teaching assistant for *Stellar Structure and Evolution* (AST220, *stellar physics*), taught by Prof. Rory Smith. I played an active role in laboratory sessions and was also responsible for grading exams and projects.

During the first semester of 2025, I served as a teaching assistant for *Computational*

Astronomy I (AST205), also taught by Prof. Rory Smith. I actively participated in each class, as the course is strongly practical and students require constant guidance to develop their projects. I was also responsible for grading the projects.

From August to the end of the second semester of 2025, I worked as a teaching assistant for *Accretion Disk Theory: Theoretical Foundations, Numerical Simulations, and Observational Connection* (AST331), also participating actively in laboratory sessions. The course was taught by Prof. Matías Montesinos.

Conferences & Workshops

II Workshop on the Halo-Galaxy Connection, UTFSM, Santiago (2022).
III Workshop on the Halo-Galaxy Connection, Universidad de Atacama, Copiapó (2023).
IV Workshop for Students in Physical and Astronomical Sciences, Universidad de La Serena, La Serena (2025).
XX SOCHIAS Annual Meeting, Universidad Austral de Chile, Puerto Montt (2025).

Presentations

The Core and Cusp Dichotomy in TNG, *III Workshop on the Halo-Galaxy Connection*, Universidad de Atacama, Copiapó (2023).

Probing the Inner Structure of Galaxies Through Dark Matter Density Profiles: Baryonic Effects and Cosmic Evolution, *IV Workshop for Students in Physical and Astronomical Sciences*, Universidad de La Serena, La Serena (2025).

Probing the Inner Structure of Galaxies Through Dark Matter Density Profiles: Baryonic Effects and Cosmic Evolution, *XX SOCHIAS Annual Meeting*, Universidad Austral de Chile, Puerto Montt (2025).

Programming Languages

Python, SQL, MATLAB, Mathematica.

Astronomical Software

Cosmology and structure formation: *IllustrisTNG*, *MUSIC*, *DICE*, *RAMSES*, *Corrfunc*

Hydrodynamics and N-body systems: *FARGO3D*, *REBOUND*

Stellar evolution: *Single Stellar Evolution*, *STATSTAR*

Spectral analysis and synthesis: *SPLAT-VO*, *MAGPHYS*, *FSPS*

Photometric and morphological analysis: *APT*, *SExtractor*, *GALFIT*

Languages

Spanish: Native
English: Advanced

Outreach and Personal Projects

Recognition: I was awarded a certificate of recognition for presenting a scientific poster at the IV Workshop for Students in Physical and Astronomical Sciences, in recognition of the quality, clarity, and scientific communication of my research.

Outreach: During my undergraduate studies, I represented my degree program in a vocational outreach discussion, aimed at high-school students interested in astronomy and the sciences.